

# Commutator

Def. Let  $G$  be a group. Given  $x, y \in G$ , define

$$[x, y] \stackrel{\text{def}}{=} x^{-1}y^{-1}xy$$

is called the commutator of  $x$  and  $y$ . And, given two subset  $A$  and  $B$ ,

$$[A, B] \stackrel{\text{def}}{=} \langle [a, b] \mid a \in A, b \in B \rangle$$

Particularly,  $G' \stackrel{\text{def}}{=} [G, G]$  is called the derived subgroup.

Prop.  $G'$  char  $G$ , (so  $G' \trianglelefteq G$ ).

Proof. Given  $\sigma \in \text{Aut}(G)$  and  $[x, y]$  for arbitrary  $x, y \in G$ ,

$$\sigma([x, y]) = \sigma(x^{-1}y^{-1}xy) = \sigma(x)^{-1} \cdot \sigma(y)^{-1} \cdot \sigma(x) \cdot \sigma(y) = [\sigma(x), \sigma(y)]$$

$$\text{So, } \sigma[[G, G]] = [G, G].$$

Prop. Given  $H \leq G$ ,  $H \trianglelefteq G$  if and only if  $[H, G] \leq H$

Proof.  $H \trianglelefteq G \Leftrightarrow$  for all  $g \in G$  and  $h \in H$ ,  $ghg^{-1} \in H$

$$[H, G] \leq H \Leftrightarrow \text{for all } g \in G \text{ and } h \in H, [h, g] = h^{-1}g^{-1}hg \in H.$$

So proved.  $\square$

Prop. Given  $N \trianglelefteq G$ ,  $G/N$  is abelian if and only if  $G' \leq N$ .

Proof.  $G/N$  is abelian iff for any  $xN, yN \in G/N$ ,  $xyN = yxN$

iff for any  $x, y \in G$ ,  $x^{-1}y^{-1}xy \in N$

iff  $G' \leq N$ .  $\square$

Note:  $G'$  is the smallest normal subgroup the quotient is abelian.

In other say,  $G/G'$  is the largest abelian quotient group.

$$G' = \bigcap_{N \trianglelefteq G, G/N \text{ is abelian}} N$$

Prop. If  $\phi: G \rightarrow A$  is a Homomorphism into an abelian group  $A$ , then  $G' \leq \ker \phi$ .

Proof. By the first Isomorphism Theorem,  $G/\ker \phi \cong \text{Im } \phi \leq A$ .

Since  $A$  is abelian, so  $\text{Im } \phi$  is also abelian. Therefore,  $G' \leq \ker \phi$ .  $\square$

Alternatively, given  $x, y \in G$ ,  $\phi([x, y]) = \phi(x)^{-1}\phi(y)^{-1}\phi(x)\phi(y) = 1_A$ .  $\square$

Homomorphism decomposition.

Let  $\varphi: G \rightarrow A$  be a Group Homomorphism into abelian group  $A$ .  
By the First-Isomorphism theorem,  $G/\ker \varphi \cong \text{Im } \varphi \leq A$ , so  $G' \leq \ker \varphi$ .  
Therefore,  $\psi: G/G' \rightarrow A; \alpha G' \mapsto \varphi(\alpha)$  is well-defined because

$$\alpha_1 G' = \alpha_2 G' \Leftrightarrow \alpha_1^{-1} \alpha_2 \in G' \Rightarrow \alpha_1^{-1} \alpha_2 \in \ker \varphi \Leftrightarrow \varphi(\alpha_1^{-1} \alpha_2) = e \Leftrightarrow \varphi(\alpha_1) = \varphi(\alpha_2)$$

Now, the onto Homomorphism,  $\pi: G \rightarrow G/G'; g \mapsto gG'$  satisfies:

$$\begin{array}{ccc} G & \xrightarrow{\varphi} & A \\ \pi \downarrow \wr & \searrow \varphi & \\ G/G' & \xrightarrow{\psi} & A \end{array} \quad \varphi = \psi \circ \pi.$$

And,

$$\ker \varphi = \varphi^{-1}[\{e\}] = \pi^{-1}[\psi^{-1}[\{e\}]] = \pi^{-1}[\ker \psi].$$

Commutator of Typical Subgroups.

$D_{2n}$ , the dihedral of order  $2n$ .

Prop.  $D'_{2n} = \langle r^2 \rangle \cong \begin{cases} C_n & \text{if } n \text{ odd} \\ C_{\frac{n}{2}} & \text{if } n \text{ even} \end{cases}$

$S_n$ , the symmetric group on  $n$  letters

Prop.  $S_n' = A_n$ .

Proof. Since  $S_n/A_n \cong C_2$ , abelian, so  $S_n' \leq A_n$ .

Debris,

$$[S_3, S_3] = \langle [a, b] \mid a, b \in S_3 \rangle$$

$$[(1\ 2), (1\ 3)] = (1\ 2)(1\ 3)(1\ 2)(1\ 3) = (1\ 3\ 2)(1\ 3\ 2) = (1\ 2\ 3)$$

$$[(1\ 2), (2\ 3)] = (1\ 2)(2\ 3)(1\ 2)(2\ 3) = (1\ 2\ 3\ 1)(1\ 2\ 3\ 1) = (1\ 3\ 2)$$

$$[(1\ 3), (2\ 3)] = (1\ 3)(2\ 3)(1\ 3)(2\ 3) = (3\ 2\ 1)(3\ 2\ 1) = (1\ 2\ 3)$$

$$[(1\ 2), (1\ 2\ 3)] = (1\ 2)(1\ 2\ 3)^{-1}(1\ 2)(1\ 2\ 3) = (1\ 2)(3\ 2\ 1)(1\ 2)(1\ 2\ 3) = (1\ 3\ 2)$$

$$[(1\ 3), (1\ 2\ 3)] = (1\ 3)(1\ 2\ 3)^{-1}(1\ 3)(1\ 2\ 3) = (1\ 3)(3\ 2\ 1)(1\ 3)(1\ 2\ 3) = (1\ 3\ 2)$$

$$[(2\ 3), (1\ 2\ 3)] = (2\ 3)(1\ 2\ 3)^{-1}(2\ 3)(1\ 2\ 3) = (2\ 3)(3\ 2\ 1)(2\ 3)(1\ 2\ 3) = (1\ 3\ 2)$$

$$[(1\ 2\ 3), (1\ 3\ 2)] = \text{id}.$$

$$\text{Therefore, } [S_3, S_3] = \langle (1\ 2\ 3), (1\ 3\ 2) \rangle = \{\text{id}, (1\ 2\ 3), (1\ 3\ 2)\} = A_3.$$

Alternatively, since  $A_3$  is normal subgroup of  $S_3$  and  $S_3/A_3 \cong C_2$ , so  $[S_3, S_3] \subseteq A_3$ . Conversely, since  $(1\ 3\ 2) \in A_3$  and  $(1\ 3\ 2) = [(1\ 2), (2\ 3)] \in [S_3, S_3]$ . Therefore  $[S_3, S_3] \supseteq A_3 = \langle (1\ 3\ 2) \rangle$ .

Since  $S_n = \langle (1\ 2), (1\ 2\ 3 \dots n) \rangle$ , because:

$$(1\ 2 \dots n)(1\ 2)(n \dots 2\ 1) = (2\ 3)$$

$$(1\ 2 \dots n)(2\ 3)(n \dots 2\ 1) = (3\ 4)$$

$$\vdots$$
$$(1\ 2 \dots n)(n-2\ n-1)(n \dots 2\ 1) = (n-1\ n)$$

and for any  $1 \leq i \leq n$

$$(i\ i+1)(i+1\ i+2)(i\ i+1) = (i\ i+2)$$

$$(i\ i+2)(i+2\ i+3)(i\ i+2) = (i\ i+3)$$

$$\vdots$$
$$(i\ i+j)(i+j\ i+j+1)(i\ i+j) = (i\ i+j+1), \text{ inductively,}$$

$\langle (1\ 2), (1\ 2 \dots n) \rangle$  contains all transpositions in  $S_n$ , so it generates  $S_n$ .

Now,

$$S_n' = [S_n, S_n] =$$

$\Gamma_{A_n}$ , the alternating group.

$\Gamma_8$ . The Quaternion.